Generating Natural Language Proofs with Verifier-Guided Search

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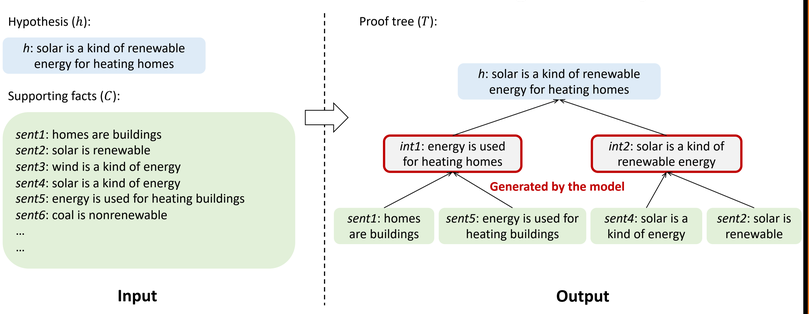
Generating Natural Language Proofs

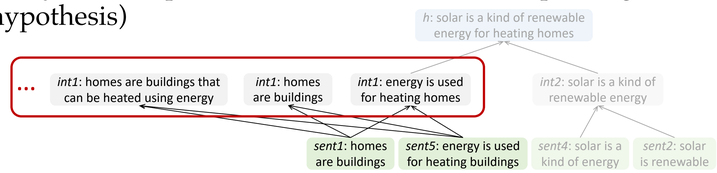
- **Input**: a hypothesis h and a set of supporting facts $C = \{sent_1, sent_2, \ldots, sent_n\}$ in natural language
- **Output**: a proof tree T for deriving h from a subset of C
  - The root node is h; the leaf nodes are sentences in C
  - Others are intermediate conclusions *generated by the model*



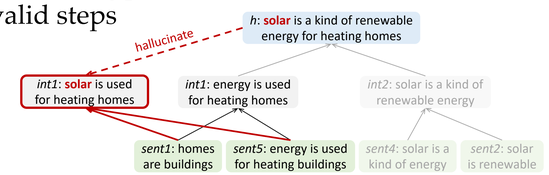
- **Single-Shot Methods**
  - Generate the entire proof altogether
- **Stepwise Methods**
  - Generate the proof step by step
  - ✓ Better leverage compositionality and generalize to longer proofs
  - ✗ Achieved limited success on challenging proofs authored by humans, e.g., EntailmentBank [Dalvi et al. 2021]

Challenges in Generating Valid and Relevant Steps

- Existing stepwise methods struggle to generate proof steps that are both valid and relevant
- Condition on premises only:
  - Many valid steps are irrelevant (not useful for proving the hypothesis)

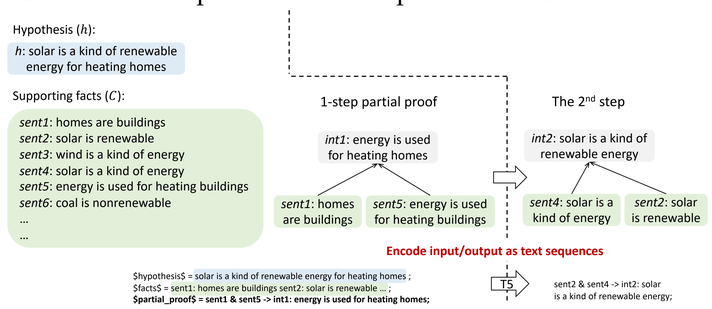


- Condition on premises + hypothesis:
  - Hallucinate invalid steps

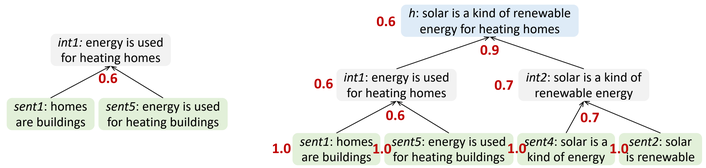


Our Method: NLProofS

- **NLProofS (Natural Language Proof Search)**
  - A new method for stepwise proof generation
  - Prover: Generate relevant steps conditioning on the hypothesis
  - Verifier: Mitigate hallucination by training an independent network to check the proof steps
  - Proof search: Use the prover/verifier to generate the final proof with the optimal validity score
- **Prover**
  - Finetune a T5 model to predict the next proof step
  - Generate multiple candidate steps via beam search

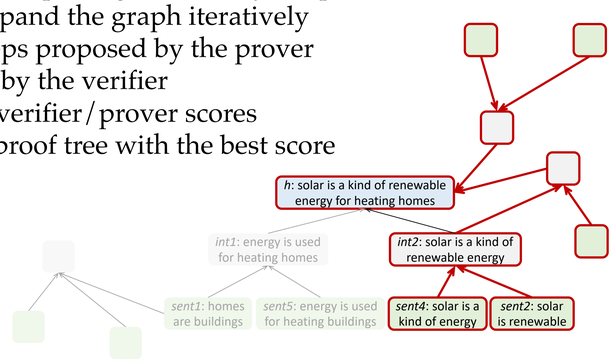


- **Verifier**
  - Input: A proof step (multiple premises, one conclusion)
  - Output: A score in [0, 1] calculated by finetuning RoBERTa
  - Step scores are aggregated to calculate proof scores



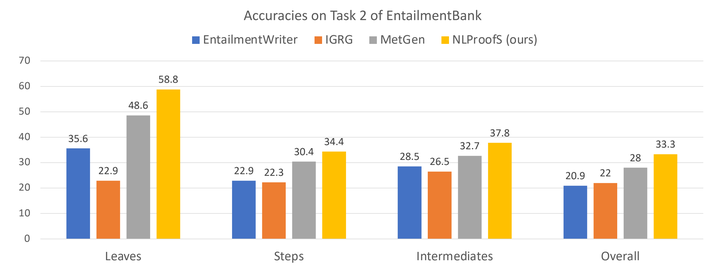
- **Proof Search**

1. Initialization: a proof generated by the prover alone
2. Iteration: expand the graph iteratively
   - Using steps proposed by the prover
   - Checked by the verifier
   - Average verifier/prover scores
3. Extraction: proof tree with the best score

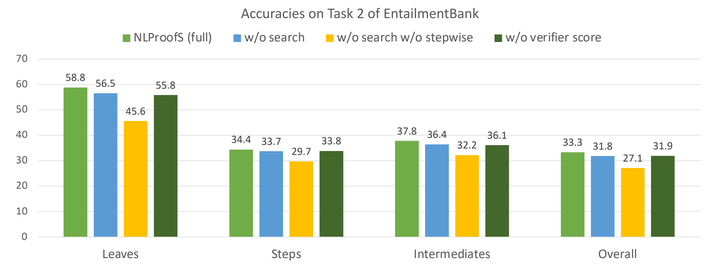


Experiments

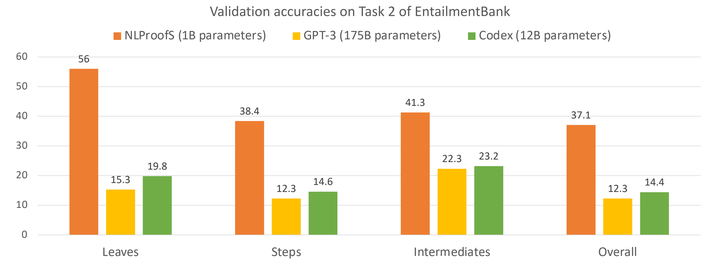
- **State-of-the-art on two benchmarks**
  RuleTaker [Tafjord et al. 2021] : Simple, synthetic profs
  EntailmentBank [Dalvi et al. 2021]: Challenging, human-written proofs
- **Main Results on EntailmentBank**
  - Single-shot baselines: EntailmentWriter [Dalvi et al. 2021]
  - Stepwise baselines: IRGR [Ribeiro et al. 2022], MetGen [Hong et al. 2022]



- **Ablations: Importance of Stepwise Verifier-Guided Search**



- **Large Language Models w/ In-Context Learning**
  - Single-shot proof generation with 7 in-context examples
  - GPT-3 and Codex cannot solve the task out of the box



1. Dalvi et al. "Explaining Answers with Entailment Trees." EMNLP 2021
2. Ribeiro et al. "Entailment Tree Explanations via Iterative Retrieval-Generation Reasoner." NAACL 2022
3. Hong et al. "METGEN: A Module-Based Entailment Tree Generation Framework for Answer Explanation." NAACL 2022
4. Tafjord et al. "ProofWriter: Generating Implications, Proofs, and Abductive Statements over Natural Language." Findings of ACL 2021